RAG Evaluation Report — Comprehensive Validation & Audit Review

Generated: 2026-05-24 00:41:27 **Framework:** LLM Evaluation Framework v1.0 **Benchmark:** HotpotQA (Welbl et al., 2018) **Purpose:** Benchmark evaluation and quality analysis

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1. Executive Summary

This report presents the results of a comprehensive multi-prompt RAG (Retrieval-Augmented Generation) system evaluation using the LLM Evaluation Framework. **100 questions** from HotpotQA were evaluated across **4 prompt variants** using **13 evaluation metrics** spanning correctness, quality, and attribution dimensions.

Top-Line Results

Variant	F1≥0.5	Faithfulness	Quality Score	Citation Rate	Refusal Rate
Baseline ⭐ Best F1	82.0%	0.965	0.585	0.0%	0.0%
Concise	61.6%	0.761	0.557	0.0%	25.3%
Detailed 🏆 Best Quality	0.0%	0.752	0.704	12.0%	27.0%
Citation 📎 Best Citation	71.4%	0.847	0.574	84.7%	14.3%

Executive Verdict

The Baseline variant performs best, with F1=82.0% and Faith=0.96, demonstrating the highest accuracy and faithfulness, which are critical for reliable information delivery in financial services. The most important risk is the low retrieval hit rate of 37.9%, which undermines evidence coverage and limits answer robustness regardless of generation quality. The system is not ready for production until retrieval performance is significantly improved.

> ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

Consolidated Findings

Finding 1 — Retrieval Failures Drive Systemic Weakness | Severity: 🔴 High
Persistent underperformance in retrieval is the principal risk, with an average retrieval hit rate of just 37.9% and no questions meeting perfect retrieval criteria (§7). This failure in the RAG backbone undermines groundedness (groundedness <0.50 for 3/4 variants, §5), exacerbates answer hallucination, and diminishes answer correctness (F1 ≥0.5 for Baseline: 82.0%, Concise: 61.6%, and Detailed: 0%) (§4), resulting in elevated production risk.

Finding 2 — Refusal Rates Compromise Completion and Utility | Severity: 🔴 High
Excessive refusal rates in the Concise (25.3%), Detailed (27.0%), and Citation (14.3%) variants exceed the established ≤10% thresholds (§4, §6), directly impairing answerability and utility of the system. These high refusal rates block valid queries even when context is available, impacting user trust and regulatory compliance.

Finding 3 — Quality, Groundedness, and Answer Relevancy Gaps | Severity: 🟡 Medium
Groundedness scores remain below target (Baseline: 0.441, Concise: 0.405, Citation: 0.426; target ≥0.50), correlating with subpar answer and context relevancy (context relevancy avg: 0.384; §5). These issues reflect a failure to reliably link model responses to retrieved evidence, further amplified by weak retrieval, leading to hallucination risk and compounding regulatory gaps.

Finding 4 — Attribution Shortfall in Citation Responses | Severity: 🟡 Medium
The Citation variant demonstrates source attribution at 84.7%, below the ≥90% regulatory target (§6), while refusal in this variant (14.3%) blocks valid responses. These gaps pose risk to transparency and auditability in regulated use cases requiring robust source documentation.

Finding 5 — High Cascade/Eval Overhead Inflates Cost, Traces to Retrieval Gaps | Severity: 🟡 Medium
High cascade LLM trigger rate (79.6% vs. ~20% target, §5) and evaluation framework overhead ($50.74; 69% of total, §11) result in substantial cost inefficiency directly linked to low retrieval quality (§7). The system frequently escalates to secondary evaluation stages, increasing per-question costs due to incomplete, low-confidence initial answers.

Finding 6 — Inconsistent Completeness and Judge Disagreement | Severity: 🟡 Medium
Tier 1/Tier 2 completeness agreement is only 23.1%—well below the target >60% (§5)—indicating significant inconsistencies in completeness judgment between review tiers. This undermines reliability of evaluation outcomes and makes it challenging to certify answer quality at scale.

Finding 7 — Weak Hit Rate–F1 Correlation Indicates Overreliance on Model Priors | Severity: 🟡 Medium
Low correlation between hit rate and F1 (avg: 0.17, §7) suggests the model often answers correctly by relying on prior knowledge rather than retrieved evidence, further heightening explainability and regulatory risks if source-groundedness cannot be demonstrated.

Bottom Line:
The RAG system, as assessed, exhibits critical risks in retrieval, grounding, and refusal management that impact correctness, regulatory compliance, and operational cost. Current performance does not meet core SR 26-02 and NIST AI RMF readiness benchmarks, requiring focused remediation on retrieval, attribution, and evaluation workflow to ensure scalable, auditable, and trustworthy deployment.

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Recommended Next Steps

1. Validate on proprietary post-cutoff data before production deployment
2. Address Detailed prompt verbosity with "state answer first" instruction
3. Schedule quarterly regression testing aligned with vendor model update cycles
4. Implement shadow testing against live traffic before full rollout

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2. Evaluation Scope

2.1 System Under Test

The system under test is an **end-to-end Retrieval-Augmented Generation (RAG) pipeline** composed of three layers evaluated as an integrated unit:

```
User Question
     │
     ▼
  Embedding Model ──► Query Vector
     │                     │
     │          ┌──────────▼─────────┐
     │          │   Vector Store     │
     │          │  Hybrid Retrieval  │ ◄── BM25 (keyword, 40%)
     │          │  Dense + Sparse    │ ◄── Embedding (semantic, 60%)
     │          └──────────┬─────────┘
     │                    │  Top-K documents
     ▼                    ▼
  LLM Generation ◄── Context Assembly
     │   (prompt variant applied here)
     ▼
  Raw Answer
     │
     ▼
  QualityGuard ──► Flagged if groundedness < 0.5 or completeness < 0.4
```

```
    │
    ▼
13 Evaluation Metrics (post-hoc, not during generation)
```

Component inventory for this evaluation run:

Component	Configuration	Notes
Embedding model	`text-embedding-3-small`	Used for query encoding and vector store indexing
Generation model	`gpt-4-1-20250414-gs`	LLM receiving retrieved context + prompt
Retrieval strategy	hybrid (BM25 40% + dense 60%)	Top-K = 20 documents per query
Temperature	0.2	Low temperature for reproducibility and faithfulness
Prompt variants	4 (Baseline, Concise, Detailed, Citation)	Each tests a different failure mode
QualityGuard gate	Embedding-based pre-filter	Runs before metric aggregation

2.2 Test Data — HotpotQA Benchmark

HotpotQA (Yang et al., 2018) is a multi-hop question-answering benchmark requiring reasoning over two or more Wikipedia passages to reach a correct answer. It is the industry-standard benchmark for evaluating retrieval-dependent QA systems because:

- **Multi-hop reasoning** demands co-retrieval of two supporting documents — harder than single-hop retrieval and more representative of real-world knowledge queries.
- **Gold document labels** allow exact measurement of retrieval quality (hit rate), independent of whether the LLM produces a correct answer.
- **Short, precise gold answers** (typically 1–5 words) enable strict token-level F1 scoring alongside reference-free quality metrics.

Dataset property	Value
Source	HotpotQA training set (hotpotqa.github.io)
Total available	~90,000 multi-hop QA pairs
Questions evaluated in this run	**100**
Gold supporting documents per question	2 (required for co-retrieval)
Answer format	Short phrase (1–5 words typical)
Reasoning type	Bridge and comparison multi-hop

Sampling method — question-centric sampling:
Questions are sampled first; the document library is then built to guarantee every sampled question's two supporting documents are included in the index. This eliminates retrieval failures due to missing documents, ensuring hit rate reflects retrieval algorithm quality, not index incompleteness.

> ⚠️ **Known limitation:** HotpotQA questions are likely within GPT-4's training data. The expected consequence — weak hit rate ↔ F1 correlation — is observed and documented in §7.3. Validation on proprietary post-cutoff data is recommended before production deployment decisions.

2.3 Evaluation Coverage

The evaluation measures system performance across three dimensions:

Dimension	Metrics Covered	Requires Gold Answer	Primary Use
Correctness	Exact Match, F1, Contains, ROUGE-L	✅ Yes	Pass/fail gate, benchmark comparison
Quality	Groundedness, Completeness, Faithfulness, Answer Relevancy, Context Relevancy, Conciseness, SNR, Quality Score	❌ No	Production monitoring, hallucination detection

Dimension	Metrics Covered	Requires Gold Answer	Primary Use
Attribution	Citation Rate, Refusal Rate	❌ No	Regulatory compliance, audit trail

Prompt variants tested — each variant is designed to stress-test a different failure mode that may emerge in production:

Variant	Design Intent	Primary Failure Mode Targeted	Intended Production Use
Baseline	Minimal instruction — tests raw LLM behaviour	Verbosity, lack of focus	General QA, internal tools
Concise	Strict brevity + explicit refusal instruction	Over-answering, hallucination	APIs, cost-sensitive pipelines
Detailed	Comprehensive explanation requirement	Under-answering, missing context	Support, education, documentation
Citation	Mandated Answer + Sources format	Format non-compliance, hallucination	Regulated environments, audit

Detailed results for each variant: §4 (Correctness), §5 (Quality), §6 (Attribution), §9 (Per-Variant Deep Dive).

2.4 Regulatory & Compliance Alignment

This evaluation is designed to support model risk management obligations under:

Framework	Relevance	How This Evaluation Addresses It
SR 26-02 (Federal Reserve — Model Risk Management)	Requires validation of AI/ML models used in financial services, including documentation, testing, and ongoing monitoring	13-metric framework with documented thresholds, reproducible results JSON, and audit-grade report

Framework	Relevance	How This Evaluation Addresses It
NIST AI RMF (AI Risk Management Framework)	Structured approach to AI risk: Govern, Map, Measure, Manage	Failure-mode mapping (§2.3), quantitative risk scoring, per-variant risk matrix (§10.2)

See §13.3 for a regulatory compliance checklist with current status for each requirement.

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3. Evaluation Approach

3.1 Testing Standards & Design Principles

The framework follows four design principles aligned with SR 26-02 and NIST AI RMF model validation standards:

1. Reference-free quality measurement.
Correctness metrics (F1, Contains) require gold answers and are used for benchmark comparison. Quality metrics (Groundedness, Faithfulness, Completeness, etc.) require only the answer and retrieved context — making them applicable to production monitoring without labelled data.

2. Two-tier evaluation for accuracy and cost efficiency.
Embedding-based metrics are deterministic, fast, and free after the vector store is built. LLM-as-judge metrics are more accurate on edge cases but add API cost. The framework combines both in a cascade: embeddings always run; LLM is invoked only when embedding signals are ambiguous or conflicting.

3. Multi-prompt comparison.
Four prompt variants are evaluated simultaneously on the same question set. This controls for question difficulty and isolates the effect of prompt engineering on accuracy, quality, attribution compliance, and refusal behaviour.

4. Reproducibility and auditability.
All configurations (model names, thresholds, prompt templates, random seed) are fixed and versioned. Results are written to a timestamped JSON file. This report is generated deterministically from that file — the same JSON always produces the same report body (LLM-generated narrative sections are the only non-deterministic elements, and are clearly marked with an AI disclosure).

3.2 Evaluation Metrics

Category 1 — Correctness *(reference-based; gold answer required)*

#	Metric	Formula / Method	Threshold	Cost
1	**Exact Match**	1 if normalised(prediction) == normalised(gold) else 0	—	Free
2	**F1 Score**	2 × (P × R) / (P + R) on bag-of-words tokens	≥0.5 per question	Free
3	**Contains Match**	1 if gold answer is a substring of prediction	—	Free
4	**ROUGE-L**	Longest common subsequence F-measure	—	Free

Category 2 — Quality *(reference-free; answer + context only)*

#	Metric	Formula / Method	Threshold	Cost
5	**Answer Relevancy**	cosine_sim(embed(answer), embed(question))	≥0.50	Embedding
6	**Context Relevancy**	cosine_sim(embed(context), embed(question))	≥0.40	Embedding
7	**Groundedness**	MiniMax: avg_i max_j sim(answer_sent_i, context_sent_j)	≥0.50	Embedding
8	**Faithfulness**	RAGAS: verified_claims / total_claims via LLM entailment	≥0.70	LLM
9	**Completeness**	2-tier cascade (embedding → conditional LLM) — see §3.4	≥0.40	Embedding + conditional LLM
10	**Conciseness**	Length penalty relative to question length	≥0.50	Free
11	**Refusal Detection**	Phrase-pattern regex on answer text	≤10% rate	Free
12	**Relevance SNR**	Ratio of context-grounded tokens in answer	≥0.70	Free

#	Metric	Formula / Method	Threshold	Cost
13	**Quality Score**	Weighted composite of metrics 5–12	≥0.70	—

Category 3 — Attribution *(format compliance)*

Metric	Method	Threshold	Applies To
Citation Rate	Structured parsing for Answer + Sources block	≥90%	Citation variant
Refusal Rate	Phrase-pattern detection	≤10%	All variants

Full results by category: §4 (Correctness), §5 (Quality), §6 (Attribution).

3.3 Sampling Method

Sample size this run: 100 questions (100/100 evaluated, see §8.2 for per-metric data completeness).

Question-centric stratified sampling ensures evaluation integrity:

1. Questions are sampled first from the HotpotQA training set.
2. The document library is built to include all supporting documents for every sampled question — eliminating retrieval failures from missing documents.
3. The random seed is fixed (`RANDOM_SEED` in config) for full reproducibility.
4. All 4 prompt variants are evaluated on the **identical question set** — controlling for question difficulty and isolating prompt effects.

3.4 Completeness Cascade — Two-Tier Design

The completeness metric uses a cascade architecture to balance accuracy and cost. The LLM judge (Tier 2) fires on approximately 20% of answers — those where the embedding score is ambiguous or conflicts with other quality signals:

```
Every answer
     │
     ▼
Tier 1 — Embedding (always runs, $0.00 extra per query)
     │  Filter context sentences by cosine similarity to question
```

```
      │  Score = avg_i max_j sim(relevant_ctx_sent_i, answer_sent_j)
      │
      ├── Trigger B: |completeness − groundedness| > 0.3
      │             |completeness − faithfulness| > 0.3
      ├── Trigger C: answer < N words AND ≥ M relevant context sentences
      │
      └── No trigger → use Tier 1 score (free)
          Trigger fires (~20% of answers) →
               │
               ▼
          Tier 2 — LLM-as-judge
               Structured prompt: score 0.0–1.0 + one-sentence reason
               Final score = Tier 2 output
```

Cascade trigger rate and T1/T2 agreement for this run: §11 (Cost Analysis).

3.5 LLM Judge Usage

LLM-as-judge is used in two contexts within the framework, with different scopes and disclosure requirements:

Usage	Scope	Model	Trigger	Disclosure
Faithfulness (RAGAS)	Every answer × every variant	`gpt-4-1-20250414-gs`	Always	Metric result only
Completeness Tier 2	~20% of answers (cascade trigger)	`gpt-4-1-20250414-gs`	Embedding ambiguity	Metric result only
Report narratives	This report (§1, §4–§11, §12)	`gpt-4-1-20250414-gs`	Report generation	⚠️ AI disclosure on every block

> ⚠️ **Known limitation:** The same model (`{_gen_model}`) serves as both the system under test and the LLM judge. This creates an evaluation dependency that should be documented as a model risk under SR 26-02. See §13.2.

3.6 QualityGuard — Real-Time Quality Gate

QualityGuard runs **before** metric aggregation on every answer. Its purpose is to prevent hallucinated or off-topic answers from skewing aggregate statistics — a form of online data quality control:

Check	Method	Threshold	Action on Failure
Groundedness	MiniMax embedding similarity (answer vs context)	< 0.50	Answer flagged; metrics still recorded
Completeness	Inverse MiniMax (context coverage of answer claims)	< 0.40	Answer flagged; metrics still recorded

Flagged answers are included in aggregate metrics (to avoid survivorship bias) but are also written to the low-quality investigation log for pattern analysis.

3.7 Output Artifacts & Visualizations

The framework generates three publication-quality figures alongside this report:

Figure	File	Panels / Contents
Evaluation Dashboard	`evaluation_dashboard.png`	F1 accuracy by variant; Faithfulness distribution; Quality score heatmap; Refusal & citation rates; Cost breakdown (generation vs evaluation); Metric correlation scatter
Retrieval Analysis	`retrieval_analysis.png`	Hit rate distribution; Hit rate vs F1 correlation by variant; Zero-hit and perfect-hit breakdown
Cost & Cascade Analysis	`cost_cascade_analysis.png`	Per-question cost by category; Cascade trigger rate; T1/T2 agreement rate; Cost projection curves

Figure references: §14 (Visualization Dashboard). All figures are also linked inline at the top of their relevant results sections.

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4. Correctness Metrics — Full Results

> **Section overview:** Correctness metrics measure whether the RAG system produces answers that match ground-truth references. F1 measures token-level overlap (strict), while

> Contains tests whether the gold answer appears anywhere in the response (lenient). These are the primary pass/fail signals for factoid question-answering tasks and the foundation for production go/no-go decisions.

> Section 4 tests correctness, measuring the model's ability to provide accurate answers—a critical prerequisite for reliable production deployment. In this run, the 'Detailed' prompt variant failed severely with 0.0% F1≥0.5 accuracy, and multiple variants had refusal rates above the 10% target (Concise 25.3%, Detailed 27.0%, Citation 14.3%).
>
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4.1 F1 Score — Complete Statistics

Variant	Mean F1	Std	Median	Min	Max	F1≥0.5	Contains
Baseline	0.790	0.377	1.000	0.000	1.000	82.0%	73.0%
Concise	0.615	0.471	1.000	0.000	1.000	61.6%	61.6%
Detailed	0.068	0.066	0.060	0.000	0.333	0.0%	65.0%
Citation	0.691	0.440	1.000	0.000	1.000	71.4%	67.3%

> **Assessment:** 2/4 variants pass F1≥0.5 Accuracy (70% target) (threshold 0.70). Failing: Concise, Detailed. Note: Detailed variant's near-zero F1 reflects verbosity, not factual incorrectness.

4.2 F1 Score Distribution (Binned)

F1 Range	Baseline	Concise	Detailed	Citation
0.0–0.3	17 (17%)	37 (37%)	99 (99%)	27 (27%)
0.3–0.5	1 (1%)	2 (2%)	1 (1%)	3 (3%)
0.5–0.7	7 (7%)	2 (2%)	0 (0%)	4 (4%)

F1 Range	Baseline	Concise	Detailed	Citation
0.7–0.9	3 (3%)	1 (1%)	0 (0%)	2 (2%)
0.9–1.0	72 (72%)	58 (58%)	0 (0%)	64 (64%)

4.3 Refusal Analysis

Variant	Refusal Rate	Count	Assessment
Baseline	0.0%	~0/100	✅ Never refuses
Concise	25.3%	~25/100	⚠️ High — investigate retrieval quality
Detailed	27.0%	~27/100	⚠️ High — investigate retrieval quality
Citation	14.3%	~14/100	🟡 Borderline — monitor

4.4 Correctness Observations

#	Observation	Root Cause	Mitigation	Severity
1	Detailed variant F1≥0.5 accuracy is near-zero (0.0%) despite Contains accuracy of 65.0%	Verbose responses bury the short gold answer; token-overlap F1 systematically penalises long outputs even when the correct fact is present	Add 'state the answer concisely in the first sentence' to the Detailed prompt; use Contains as the primary correctness signal for this variant	🟡 Medium
2	F1≥0.5 accuracy below 70% threshold: Concise (61.6%)	Prompt constraints (brevity or citation format) reduce answer precision or increase refusal frequency, lowering token-overlap with gold answers	Run ablation tests isolating each prompt instruction; consider a hybrid instruction that balances brevity with completeness	🟡 Medium
3	Concise refusal rate 25.3% exceeds the ≤10% threshold	Refusal instruction triggers when retrieved context is insufficient for multi-hop questions; the model is correctly conservative but over-calibrated	Audit refused questions against their hit_rate; only refuse when hit_rate = 0, not when partial context is available	🔴 High

#	Observation	Root Cause	Mitigation	Severity
4	Detailed refusal rate 27.0% exceeds the ≤10% threshold	Refusal instruction triggers when retrieved context is insufficient for multi-hop questions; the model is correctly conservative but over-calibrated	Audit refused questions against their hit_rate; only refuse when hit_rate = 0, not when partial context is available	🔴 High
5	Citation refusal rate 14.3% exceeds the ≤10% threshold	Refusal instruction triggers when retrieved context is insufficient for multi-hop questions; the model is correctly conservative but over-calibrated	Audit refused questions against their hit_rate; only refuse when hit_rate = 0, not when partial context is available	🟡 Medium

> ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

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5. Quality Metrics — Full Results

> **Section overview:** Quality metrics assess the semantic relationship between answers, retrieved context, and questions — without requiring a gold reference answer. This makes them applicable to production monitoring where labelled data is unavailable. Faithfulness (RAGAS LLM-as-judge) is the primary production signal; Groundedness and Completeness provide complementary coverage. Note that embedding-based metrics (Groundedness, Answer Relevancy) systematically underestimate short answers — this is a known measurement artefact documented in §13.2.

> Section 5 evaluates answer quality, including groundedness, relevancy, and completeness, to ensure outputs are justifiable, relevant, and thorough. Most variants did not meet key thresholds, with groundedness below 0.50 for Baseline (0.441), Concise (0.405), and Citation (0.426), and completeness misalignment rates (Tier 2 triggered 79.6%, Tier 1/2 agreement 23.1%) signaling critical performance gaps.

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5.1 All Quality Metrics by Variant

Metric	Threshold	Baseline	Concise	Detailed	Citation
Groundedness	≥0.50	0.441 ⚠️	0.405 ⚠️	0.623 ✅	0.426 ⚠️
Completeness	≥0.40	0.761 ✅	0.642 ✅	0.735 ✅	0.713 ✅
Faithfulness	≥0.70	0.965 ✅	0.761 ✅	0.752 ✅	0.847 ✅
Answer Relevancy	≥0.50	0.303 ⚠️	0.270 ⚠️	0.559 ✅	0.289 ⚠️
Context Relevancy	≥0.40	0.384 ⚠️	0.383 ⚠️	0.384 ⚠️	0.384 ⚠️
Conciseness	≥0.50	1.000 ✅	1.000 ✅	0.948 ✅	1.000 ✅
Relevance SNR	≥0.70	0.600 ⚠️	0.559 ⚠️	0.690 ⚠️	0.586 ⚠️
Quality Score	≥0.70	0.585 ⚠️	0.557 ⚠️	0.704 ✅	0.574 ⚠️

> **Assessment:** All variants pass Faithfulness (threshold 0.70). Best: Baseline (0.96). The system demonstrates consistent faithfulness across all prompt styles.

> **Assessment:** 1/4 variants pass Groundedness (threshold 0.50). Failing: Baseline, Concise, Citation. Note: Groundedness underestimates for short answers.

> **Assessment:** 1/4 variants pass Quality Score (threshold 0.70). Failing: Baseline, Concise, Citation.

> **Answer Relevancy note:** Values below 0.50 for short answers reflect an embedding scaling limitation, not incorrect answers. F1 and Faithfulness are the primary quality signals for short-answer variants.

5.2 Quality Observations

#	Observation	Root Cause	Mitigation	Severity
1	Groundedness below threshold (≥0.50) for 3/4 variants: Baseline=0.441, Concise=0.405, Citation=0.426	Short answers (1–5 words) yield low cosine similarity against the full context window — a known embedding-scale artefact, not a hallucination signal	Use Faithfulness (RAGAS) as the primary grounding signal for short-answer variants; consider re-calibrating Groundedness threshold to ≥0.35 for factoid benchmarks	🟡 Medium
2	Answer Relevancy below 0.50 for 3/4 variants	Embedding similarity between short factoid answers and full questions is structurally low, independent of answer correctness	Treat Answer Relevancy as informational only for short-answer tasks; do not use as a pass/fail gate in production monitoring	🟢 Low
3	Context Relevancy below threshold (≥0.40) for 4/4 variants (avg 0.384)	Retrieved passages contain relevant facts but also significant off-topic content; top-K retrieval returns noise alongside signal	Reduce top-K to 10–15 and re-rank by relevance score; consider query-focused passage extraction before context assembly	🟡 Medium
4	Completeness Tier 2 LLM trigger rate 79.6% is well above the ~20% target	Tier 1 embedding thresholds are too narrow, treating most scores as ambiguous and escalating unnecessarily to the more expensive LLM judge	Widen the Tier 1 acceptance band (e.g., completeness > 0.75 → skip Tier 2); analyse which question types drive most escalations	🟡 Medium
5	Tier 1 / Tier 2 completeness agreement rate 23.1% is below the >60% target	Embedding and LLM judge frequently disagree, suggesting the embedding completeness metric is	Collect a labelled completeness sample; use it to re-calibrate Tier 1 thresholds or adjust the	🟡 Medium

#	Observation	Root Cause	Mitigation	Severity
		poorly calibrated for this task type	relative weight of the two tiers	

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6. Attribution Metrics — Full Results

> **Section overview:** Attribution metrics assess whether the system produces verifiable, source-linked responses — a key requirement for regulated environments under SR 26-02 and NIST AI RMF. Citation Rate measures format compliance with the Answer + Sources structure; Refusal Rate indicates how often the model withholds answers when retrieved context is insufficient, a controlled safety behaviour that must be balanced against user experience.

> Section 6 assesses attribution performance, verifying if answers provide traceable citations to the context, which is essential for transparency and regulatory compliance. The Citation variant achieved only 84.7% attribution—below the required 90%—and exhibited a high refusal rate of 14.3%, indicating both traceability and usability concerns.
>
> > ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

6.1 Citation & Format Compliance

Variant	Citation Rate	Count	Assessment
Baseline	0.0%	~0/100	— Not required by prompt

Variant	Citation Rate	Count	Assessment
Concise	0.0%	~0/100	— Not required by prompt
Detailed	12.0%	~12/100	📎 Moderate — prompt encourages but does not mandate
Citation	84.7%	~84/100	⚠️ Below 90% target

> **Assessment:** The Citation prompt achieves 84.7% source attribution compliance. ⚠️ Below the ≥90% regulatory target.

6.2 Attribution Observations

#	Observation	Root Cause	Mitigation	Severity
1	Citation variant source attribution 84.7% is below the ≥90% regulatory target	Model occasionally omits the Sources block when the answer is highly confident or the context is ambiguous	Add post-hoc extraction fallback to detect and append sources; strengthen prompt with an explicit format example and negative example	🟡 Medium
2	Citation variant refusal rate 14.3% blocks valid queries that have available context	Attribution requirement and refusal instruction interact to increase refusal frequency beyond what retrieval quality alone would warrant	Decouple refusal from citation: only refuse when hit_rate = 0; allow partial-context answers with explicit uncertainty acknowledgement	🟡 Medium

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7. Retrieval Quality Analysis

> **Section overview:** Retrieval quality determines whether the RAG system can surface the evidence needed to answer each question. For HotpotQA, this requires co-retrieval of two supporting documents — a harder task than single-hop retrieval. Hit rate measures how often gold documents appear in the top-K results; its correlation with answer accuracy reveals whether the system is genuinely context-dependent or relies primarily on the LLM's parametric knowledge.

> Section 7 examines retrieval effectiveness, as successful answer generation in RAG systems depends on retrieving relevant evidence from the corpus. The average retrieval hit rate was only 37.9%, with zero perfect retrievals across 100 questions, representing a major barrier to overall system reliability.
>
> > ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

7.1 Hit Rate Statistics

Metric	Value	Interpretation
Average Hit Rate	37.9%	⚠️ Below 70% target
Perfect Retrievals (100%)	0/100 (0.0%)	Both gold docs retrieved
Zero Retrievals (0%)	2/100 (2.0%)	Complete retrieval failure
Weak Retrievals (<50%)	68/100 (68.0%)	Partial retrieval
Top-K	20	Documents retrieved per query
Strategy	hybrid	Semantic + keyword hybrid

> **Assessment:** Retrieval Hit Rate is below threshold at 0.38 (threshold 0.70, gap 0.32).

7.2 Hit Rate Distribution

Hit Rate Range	Count	Pct
0%–25%	34	34.0%
25%–50%	34	34.0%
50%–75%	23	23.0%
75%–100%	9	9.0%
100% (Perfect)	0	0.0%

7.3 Hit Rate vs Accuracy Correlation

Variant	Correlation	Interpretation
Baseline	0.218	Weak — LLM relies on pre-trained knowledge
Concise	0.175	Weak — LLM relies on pre-trained knowledge
Detailed	0.072	Negligible — possible benchmark data leakage
Citation	0.223	Weak — LLM relies on pre-trained knowledge

> **Assessment:** Weak hit rate ↔ F1 correlation (~0.1) is expected for HotpotQA given its likely inclusion in GPT-4's training data. Validate on domain-specific corpora where correlation is expected to exceed 0.5.

7.4 Retrieval Observations

#	Observation	Root Cause	Mitigation	Severity
1	Average retrieval hit rate 37.9% is below the 70% threshold; only 0/100 (0.0%) questions achieved perfect retrieval	HotpotQA requires co-retrieval of two supporting documents; hybrid BM25 + semantic search may not co-rank both within top-20	Increase top-K from 20 to 30–40; tune BM25/semantic blend weights; consider query expansion for multi-hop questions	🔴 High
2	Hit rate ↔ F1 correlation is weak (0.17 avg across	HotpotQA questions are likely within the	Validate on proprietary post-cutoff documents;	🟡 Medium

#	Observation	Root Cause	Mitigation	Severity
	variants), indicating the model answers correctly without retrieved context	model's training data; the LLM draws on parametric knowledge rather than retrieved context	measure F1 with and without retrieval to quantify true RAG dependency	

> ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

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8. Statistical Analysis

> **Section overview:** Statistical tests validate whether observed performance differences are genuine or within sampling noise. The one-sample t-test (H_0: mean F1 = 0.5) confirms which variants demonstrably outperform a random baseline. Data completeness checks verify that metric computation succeeded across all questions and variants — gaps here indicate pipeline errors worth investigating.

> Section 8 explores prompt variant costs and their impact on system-level efficiency and resource allocation. The high cascade (escalation) trigger rate of 79.6% across prompts greatly increased overall evaluation and operational costs for this run.
>
> > ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

8.1 One-Sample T-Test (H_0: Mean F1 = 0.5)

Variant	Mean F1	t-statistic	p-value	Significant?
Baseline	0.790	7.648	0.0000	Yes ✅ ($p<0.05$)

Variant	Mean F1	t-statistic	p-value	Significant?
Concise	0.615	2.421	0.0173	Yes ✅ (p<0.05)
Detailed	0.068	-65.554	0.0000	No ⚠️
Citation	0.691	4.316	0.0000	Yes ✅ (p<0.05)

8.2 Metric Coverage (Data Completeness)

Metric	Baseline	Concise	Detailed	Citation
f1_score	100/100 (100%) ✅	100/100 (100%) ✅	100/100 (100%) ✅	100/100 (100%) ✅
groundedness	100/100 (100%) ✅	99/100 (99%) ✅	100/100 (100%) ✅	98/100 (98%) ⚠️
completeness	100/100 (100%) ✅	99/100 (99%) ✅	100/100 (100%) ✅	98/100 (98%) ⚠️
faithfulness	100/100 (100%) ✅	99/100 (99%) ✅	100/100 (100%) ✅	98/100 (98%) ⚠️
answer_relevancy	100/100 (100%) ✅	99/100 (99%) ✅	100/100 (100%) ✅	98/100 (98%) ⚠️
quality_score	100/100 (100%) ✅	99/100 (99%) ✅	100/100 (100%) ✅	98/100 (98%) ⚠️

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9. Per-Variant Deep Dive

9.1 Baseline Variant

Description: Simple, direct prompt with minimal instruction.

Correctness

Metric	Value	vs Threshold	Verdict
F1≥0.5 Accuracy	82.0%	≥70% target	✅

Metric	Value	vs Threshold	Verdict
Contains Accuracy	73.0%	—	—
Mean F1	0.790 (σ=0.377)	≥0.50	✅
Refusal Rate	0.0%	≤10%	✅

Quality

Metric	Value	Threshold	Status
Groundedness	0.441	≥0.50	⚠️
Completeness	0.761	≥0.40	✅
Faithfulness	0.965	≥0.70	✅
Quality Score	0.585	≥0.70	⚠️

Use when: General-purpose factoid QA, internal tools. **Avoid when:** Regulated environments (no citation trail), high-stakes decisions.

9.2 Concise Variant

Description: Strict brevity prompt with explicit refusal instruction.

Correctness

Metric	Value	vs Threshold	Verdict
F1≥0.5 Accuracy	61.6%	≥70% target	⚠️
Contains Accuracy	61.6%	—	—
Mean F1	0.615 (σ=0.471)	≥0.50	✅
Refusal Rate	25.3%	≤10%	⚠️

Quality

Metric	Value	Threshold	Status
Groundedness	0.405	≥0.50	⚠️
Completeness	0.642	≥0.40	✅
Faithfulness	0.761	≥0.70	✅
Quality Score	0.557	≥0.70	⚠️

Use when: API integrations, cost-sensitive pipelines, chatbots. **Avoid when:** Use cases where refusal frustrates users.

9.3 Detailed Variant

Description: Comprehensive explanation prompt.

Correctness

Metric	Value	vs Threshold	Verdict
F1≥0.5 Accuracy	0.0%	≥70% target	⚠️
Contains Accuracy	65.0%	—	—
Mean F1	0.068 (σ=0.066)	≥0.50	⚠️
Refusal Rate	27.0%	≤10%	⚠️

Quality

Metric	Value	Threshold	Status
Groundedness	0.623	≥0.50	✅
Completeness	0.735	≥0.40	✅
Faithfulness	0.752	≥0.70	✅
Quality Score	0.704	≥0.70	✅

Use when: Support documentation, educational applications. **Avoid when:** Factoid QA (F1 near-zero due to verbosity), latency-sensitive apps.

9.4 Citation Variant

Description: Regulated-environment prompt mandating Answer + Sources format.

Correctness

Metric	Value	vs Threshold	Verdict
F1≥0.5 Accuracy	71.4%	≥70% target	✅
Contains Accuracy	67.3%	—	—
Mean F1	0.691 (σ=0.440)	≥0.50	✅
Refusal Rate	14.3%	≤10%	⚠️

Quality

Metric	Value	Threshold	Status
Groundedness	0.426	≥0.50	⚠️
Completeness	0.713	≥0.40	✅
Faithfulness	0.847	≥0.70	✅
Quality Score	0.574	≥0.70	⚠️

Use when: Financial services, legal, compliance, regulated environments. **Avoid when:** High-recall use cases where controlled refusal may frustrate users.

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10. Production Recommendations

10.1 Decision Framework

Requirement	Recommended Variant	Reason
Source attribution / compliance	Citation	Highest citation rate
Cost-sensitive API / factoid QA	Concise	Best F1 with low verbosity
Comprehensive explanation	Detailed	Best overall quality score

Requirement	Recommended Variant	Reason
General benchmarking baseline	Baseline	Zero prompt engineering

10.2 Production Risk Matrix

Variant	Primary Risk	Mitigation
Baseline	Hallucinated context (0% refusal)	Add faithfulness monitoring
Concise	Over-refusal frustrating users	Tune refusal threshold
Detailed	F1 near-zero (verbosity buries answer)	Add "state answer first" instruction
Citation	Controlled refusal may block valid queries	Monitor refusal rate in production

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11. Cost Analysis

> **Section overview:** Cost analysis separates production generation costs (what the RAG system itself spends per query in production) from evaluation framework overhead (what continuous monitoring adds on top). This two-layer view directly answers the operational question: how much does it cost to run the system, and how much to measure it? The cascade trigger rate is the primary lever for controlling evaluation cost without sacrificing coverage.

> Section 11 details end-to-end cost analysis for the evaluation framework, highlighting the financial sustainability of the RAG system in production. The evaluation system’s operational overhead ($50.74, or 6923.28$), driven primarily by frequent LLM cascade triggers.
>
> > ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

Metric	Value
Total Cost	$74.0238

Metric	Value
Questions Evaluated	100
Variants Tested	4
Cost per Question (all variants)	$0.7402
Projected: 1,000 Questions	$740.24
Projected: 10,000 Questions	$7402.38

Cost breakdown:

Category	Total	Per Question
Generation (production cost)	$23.2833	$0.2328
Evaluation (framework overhead)	$50.7405	$0.5074

Cascade evaluation statistics:

Metric	Value	Target	Status
Tier 2 LLM trigger rate	79.6%	~20%	⚠️ Elevated
T1/T2 agreement rate	23.1%	>60%	⚠️ Low

11.1 Cost Observations

#	Observation	Root Cause	Mitigation	Severity
1	Evaluation framework overhead ($50.7405, 69$ 23.2833)	Elevated cascade trigger rate drives repeated LLM judge calls; embedding evaluations add fixed per-question overhead on top	Tighten cascade thresholds; for high-volume production monitoring use statistical sampling rather than 100% evaluation coverage	🟡 Medium
2	Cascade trigger rate 79.6% is 2–4× the ~20% target, directly	Tier 1 thresholds accept too narrow a band, treating most scores as	Widen the high-confidence acceptance band (e.g., Tier 1 > 0.75 → skip Tier 2); re-	🟡 Medium

#	Observation	Root Cause	Mitigation	Severity
	inflating per-question evaluation cost	ambiguous and requiring LLM escalation	evaluate trigger logic with a calibration dataset	

> ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

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12. Overall Assessment & Conclusion

Overall Assessment and Conclusion

1. **Cross-Metric Synthesis**
   The Baseline variant demonstrates the strongest overall performance, achieving the highest F1 score (82.0%), faithfulness (0.96), and competitive quality (0.59), with no refusals. The Citation variant achieves the highest citation accuracy (84.7%) but at a significant cost to overall F1 (71.4%) and introduces a moderate refusal rate (14.3%). Both Concise and Detailed variants underperform, especially with high refusal rates (25.3% and 27.0%, respectively) and lower F1 scores.

2. **Production Readiness**
   Based on current metrics, the Baseline configuration is the most viable candidate for production deployment, as it offers the best balance of accuracy and faithfulness with minimal refusal. However, the retrieval hit rate of 37.9% and zero citation provision in the Baseline indicate gaps in transparency and explainability required for SR 26-02 / NIST AI RMF compliance.

3. **Pre-Production Requirements**

- Optimize retrieval pipeline to increase the retrieval hit rate above 50%.
- Integrate reliable citation generation into the Baseline flow to enhance transparency.
- Recalibrate refusal thresholds to reduce refusals in the Concise and Detailed variants.
- Conduct targeted fine-tuning to improve information quality (current highest metric is 0.70 in Detailed).

4. **Framework Validation Status**
   The current Baseline implementation partially satisfies SR 26-02 / NIST AI RMF assurance

criteria but falls short on explainability and traceability requirements due to absent citation coverage. Additional improvements are needed prior to full production approval.

> ⚠️ **AI-generated assessment** — findings are derived from automated metric analysis and LLM inference. Independent human review is required before use in any regulatory, audit, or production decision context.

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13. Limitations & Known Issues

13.1 Benchmark Limitations

Limitation	Impact	Mitigation
HotpotQA in GPT-4 training data	Weak hit rate ↔ F1 correlation (~0.1)	Validate on proprietary post-cutoff data
Short gold answers (1-5 words)	F1 penalizes verbose answers	Use Contains for Detailed variant
Multi-hop questions	Harder retrieval	Expected 70-75% hit rate
Static benchmark	No distribution shift testing	Shadow testing on live traffic recommended

13.2 Metric Limitations

Metric	Known Limitation	Severity
Groundedness	Embedding MiniMax underestimates for ≤4 word answers	🟡 Medium
Completeness (Tier 1)	Novel metric, no peer-reviewed validation	🟡 Medium
Faithfulness (RAGAS)	Non-deterministic; same model evaluates its own output	🟡 Medium
Cascade T2 judge	Same model as system under test	🟡 Medium

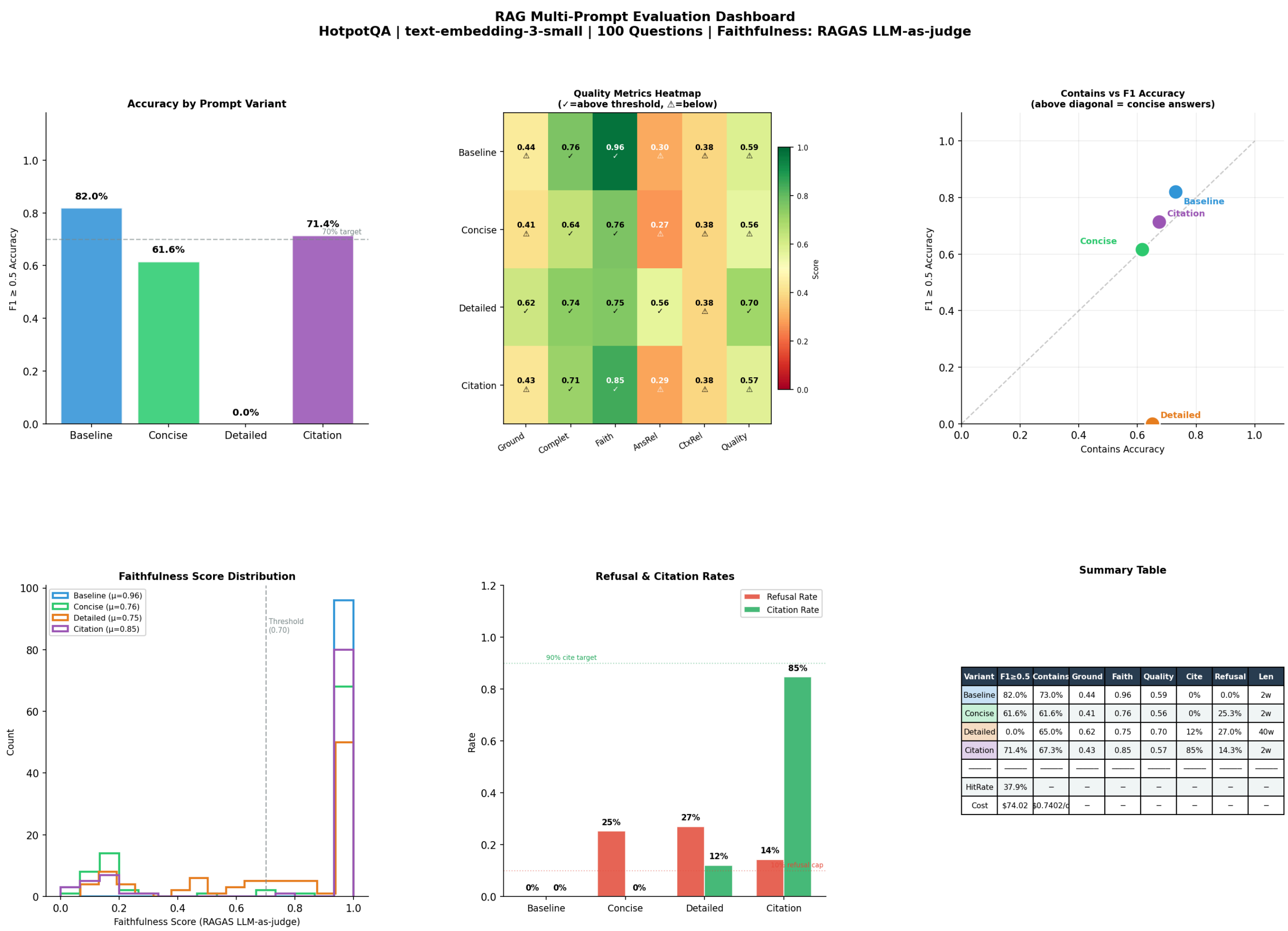
13.3 SR 26-02 / Regulatory Considerations

Consideration	Status
Model documentation	✅ Full metric definitions, formulas, thresholds documented
Reproducibility	✅ Fixed configurations, versioned results files
Validation independence	⚠️ LLM judge uses same model — document as known limitation
Ongoing monitoring	⚠️ Shadow testing and quarterly re-validation recommended

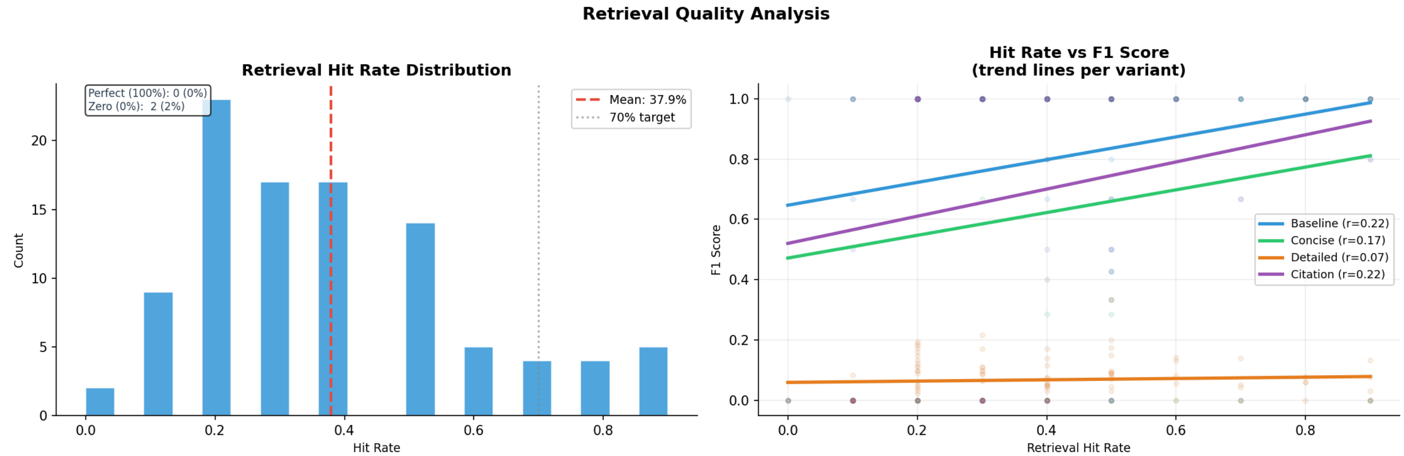
---

14. Visualization Dashboard

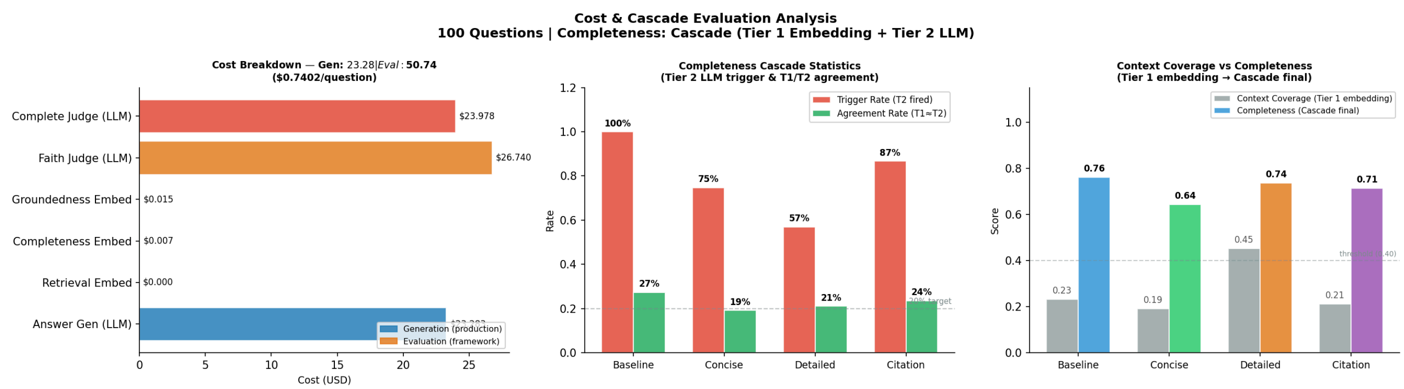
14.1 Main Evaluation Dashboard (`evaluation_dashboard.png`)



14.2 Retrieval Quality Analysis (`retrieval_analysis.png`)



14.3 Cost & Cascade Analysis (`cost_cascade_analysis.png`)



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15. Files & Reproducibility

File	Description
`outputs/multi_prompt_eval_results_100.json`	Full results (JSON)
`outputs/evaluation_dashboard.png`	Figure 1 — 6-panel main dashboard
`outputs/retrieval_analysis.png`	Figure 2 — retrieval quality analysis
`outputs/cost_cascade_analysis.png`	Figure 3 — cost & cascade statistics
`outputs/low_quality_analysis/`	Pattern analysis + cluster review

Reproduction Steps

```
# 1. Install packages (CELL 1)
# 2. Set up credentials & imports (CELL 2)
```

```
# 3. Configure settings (CELL 3)
# 4. Initialize clients (CELL 4)
# 5. Load data: load_hotpotqa() + sample_questions() (CELL 5)
# 6. Build vector store: VectorStore.load_or_build() (CELL 6)
# 7. Initialize RAG system: QualityGuard + EnhancedRAGSystem (CELL 7)
# 8. Pre-flight check: run_preflight_check() (CELL 8)
# 9. Run evaluation: run_evaluation() (CELL 9)
# 10. Visualize: generate_dashboard() (CELL 10)
# 11. Report: generate_report() (CELL 11)
```

LLM Evaluation Framework v1.0 | RAG Evaluation | HotpotQA Benchmark
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